Off the Grid

Louisville, Kentucky

2023

1087-1

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Team Interpretation of the Design Challenge and Explanation of the style and merits of the solution

The Democratic Republic of the Congo (DRC) was chosen because of its forested biome which meets the theme requirement for Off the Grid in 2023. The DRC was selected because it has a consistently warm climate year-round with plenty of sun and water which are essential for growing crops. Additionally, the DRC has ongoing issues with "armed groups [subjecting] Congolese civilians to widespread rape and sexual violence, massive human rights violations, and extreme poverty" (Council on Foreign Relations). This homestead has plenty of farmland that can be used to not only support and sustain the family that lives there, but additional crops can be sold at local markets to help revitalize the economy. With an increase in farming in the country, it can help to both reduce and stabilize food prices (Eastern Congo Initiative). The goal of this team's project was to help a family be able to escape violence and to be self-sustaining and one day overproducing to assist the local community.

The land selected is on the Lendu Plateau and has a river that runs through the property. The river has a dam built in it which serves as a source of hydroelectric power and a water source for crops during non-rainy seasons. The property has a living space opposite of the river to help protect the home against flooding. The home is made of recycled shipping containers, so it will be insulated with spray foam insulation and have solar panels that will provide electricity to the home. Additionally, the home will have a geothermal unit to help keep the house at a moderate temperature.

The shipping containers were chosen because the team has seen that these can be easily upcycled into homes. The design has lots of windows to allow for natural light, but these windows also have screens so that the home can be opened up to fresh air without the fear of

insects easily getting into the home. This is important because there are a lot of insects, such as mosquitoes, that are known for spreading diseases.

The home also has a shed that is made of half of a recycled shipping container. This will be used for storage of farming equipment and for incubators. Additionally, there is a chicken coop and a goat house. The family will be able to have three or four goats to provide a source of dairy and one of two meat sources. Goats were chosen over cattle because they require less space to healthily maintain. The goat milk can be used for drinking, or it can be used to make cheese. There will also be a chicken coop. The family will be able to raise chickens for a food source and also be able to sell eggs at the market.

The family will be rotating crops throughout the year. This will help to keep the soil healthy by not drawing too many nutrients from the soil without replenishing them. There are many crops that the property to grow, whether it be to sell or use for themselves. These crops include coffee, palm oil, rubber, cotton, sugar, tea, cocoa, coffee, cassava, plantains, maize, groundnuts, and rice (International Trade Administration).

Country's Environment, Climate, Natural Resources, and Economy

The DRC is a country located on the continent of Africa. In 1960, the DRC became independent after being under the control of Belgium. The DRC was known as the Republic of Zaire between 1971 and 1997 (Bernd Michael Wiese and Ntsomo Payanzo). It has a forested environment, tropical climate, numerous natural resources, and a vulnerable economy. The country has a population of approximately 111,860,000, and the capital is Kinshasa.

Environment

The DRC, sits on the equator and is bordered by nine countries including Angola, Burundi, Central African Republic, Republic of the Congo, Rwanda, Sudan, Tanzania, Uganda, and Zambia (Country Reports). The country is home to the Congo River which is the second longest river in the world at 2,900 miles long (Country Reports). Additionally, the DRC is a landlocked nation except for a small coastline on the Atlantic Ocean (Bernd Michael Wiese and Ntsomo Payanzo).

The topographical area of the DRC, "features a large river basin, a major valley, high plateaus, three mountain ranges, and a low coastal plain" (Bernd Michael Wiese and Ntsomo Payanzo). The highest point of the DRC is located at 2,296 feet, "in the hills of Movayi-Mbongo and Zongo in the north," and the lowest point is at 1,109 feet in Lake Mai-Ndombe (Bernd Michael Wiese and Ntsomo Payanzo). The average elevation of the DRC is 1,700 feet (Bernd Michael Wiese and Ntsomo Payanzo).

The DRC has a number of hazards and environmental issues. People in the southern region of the DRC are subject to periodic droughts, and those in the east live near active volcanoes. There is also the potential for the Congo River to flood during the rainy season. The

environmental issues include poaching of wildlife populations, deforestation, water pollution, soil erosion, and excessive mining of the mineral coltan (Country Reports).

Natural Resources

The agricultural products of the DRC are "cassava, plantains, sugar cane, maize, oil palm fruit, rice, roots/tubers nes, bananas, sweet potatoes, and groundnuts" (Central Intelligence Agency). The industries of the DRC include "mining (copper, cobalt, gold, diamonds, coltan, zinc, tin, and tungsten), mineral processing, consumer products (textiles, plastics, footwear, cigarettes), metal products, processed foods and beverages, timber, cement, commercial ship repair" (Central Intelligence Agency).

Climate

The DRC is a tropical country. In the equatorial river basin, it is hot and humid; it is, "cooler and drier in the southern highlands", and the eastern highlands are wetter and cooler (Central Intelligence Agency). On each side of the Equator there are two seasons; on the north side it is the wet season in April-October and the dry season in December-February; on the south side the wet season is in November-March and the dry season is in April-October (Central Intelligence Agency).

Economy

DRC's wealth suffers from a lot of things. Most people in DRC do not benefit from the natural resources there. DRC has a long history of conflict, authoritarian rule, political upheaval, and instability. All of these have led to an ongoing humanitarian crisis. Because of all of this, DRC is one of the poorest countries in the world. In the DRC nearly 64 percent of the population lived on less than $2.15 a day in 2021 (World Bank). Approximately, "one out of six people living in extreme poverty in [Sub-Saharan Africa] lives in DRC" (World Bank).

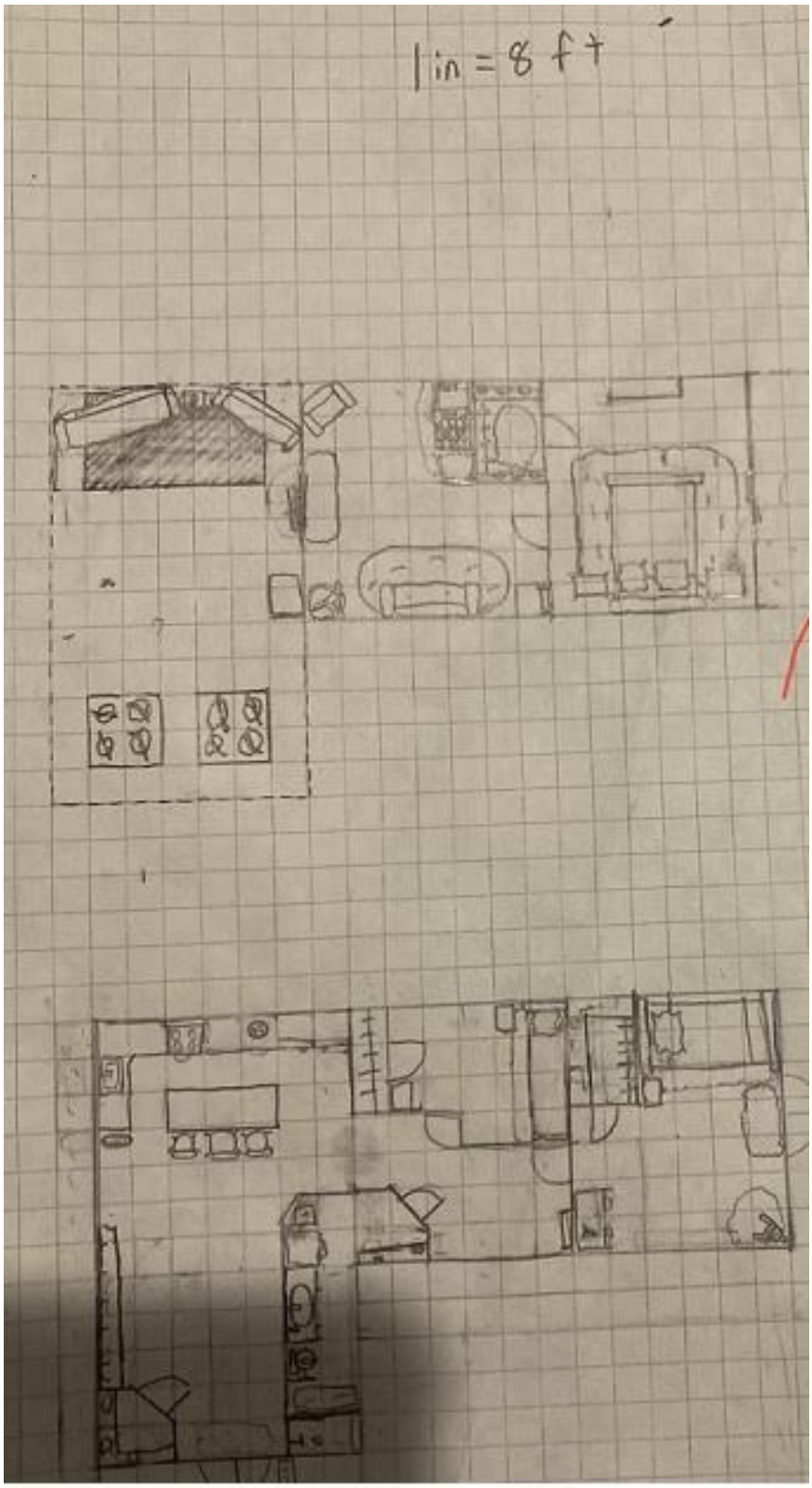
The economy in the DRC is holding steady with growth in 2022, at 6.1 percent which is on par with the 6.2 percent from 2021 (World Bank).

DRC has one of the biggest infrastructure gaps in the world. Many reforms are needed to aid the economy, but the most, “immediate challenge is to strengthen security and maintain political and macroeconomic stability while stepping up ongoing reforms to ensure sustainable growth” (World Bank). Things are looking positive because the last transfer of power was peaceful as opposed to violent coups that have plagued the country.

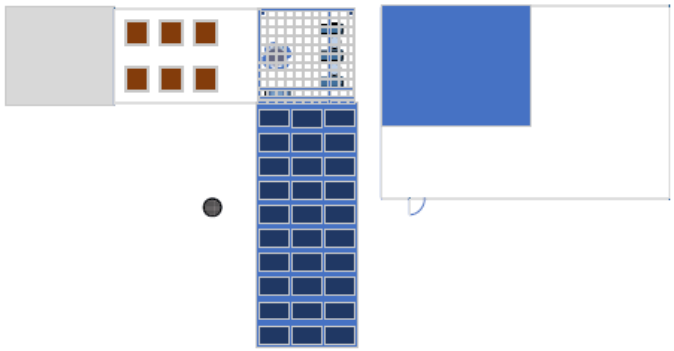
Building Materials

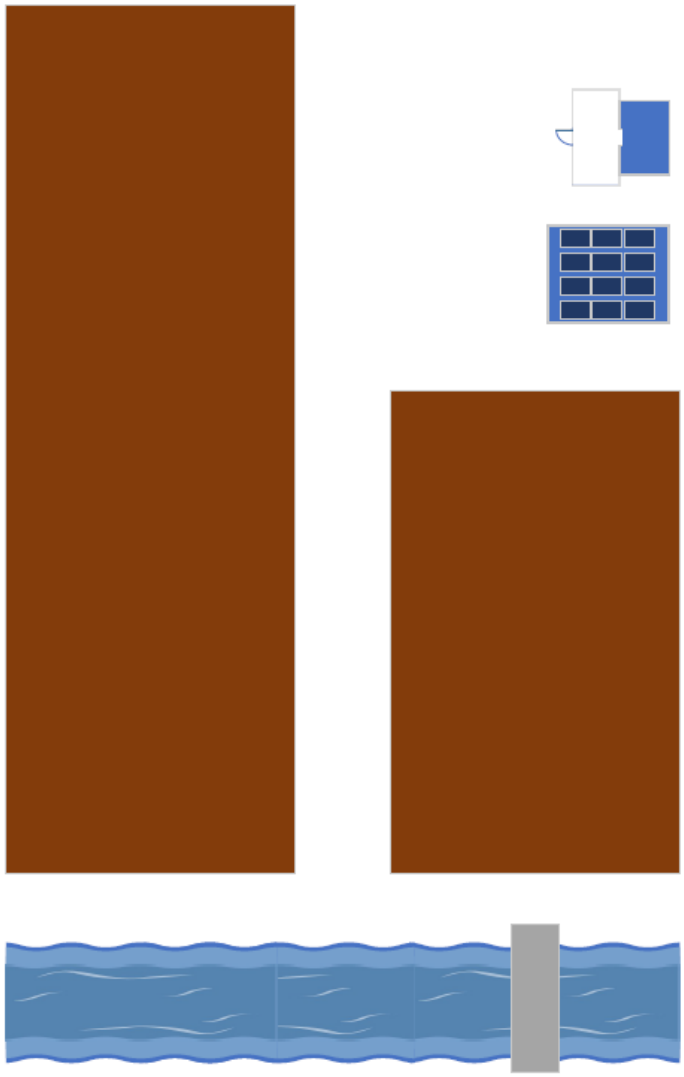
- Shipping Containers
- Sliding Glass Doors
- Triple-pane Windows
- Geothermal System
- Screen Doors
- Drywall
- Spray foam insulation
- Gray water toilets
- Iroko wood
- Granite
- Paint
- Nails
- Shingles for the goat house and chicken coop
- Eco-friendly faucets
- Pex and PVC pipes
- Cement

Floor Plans

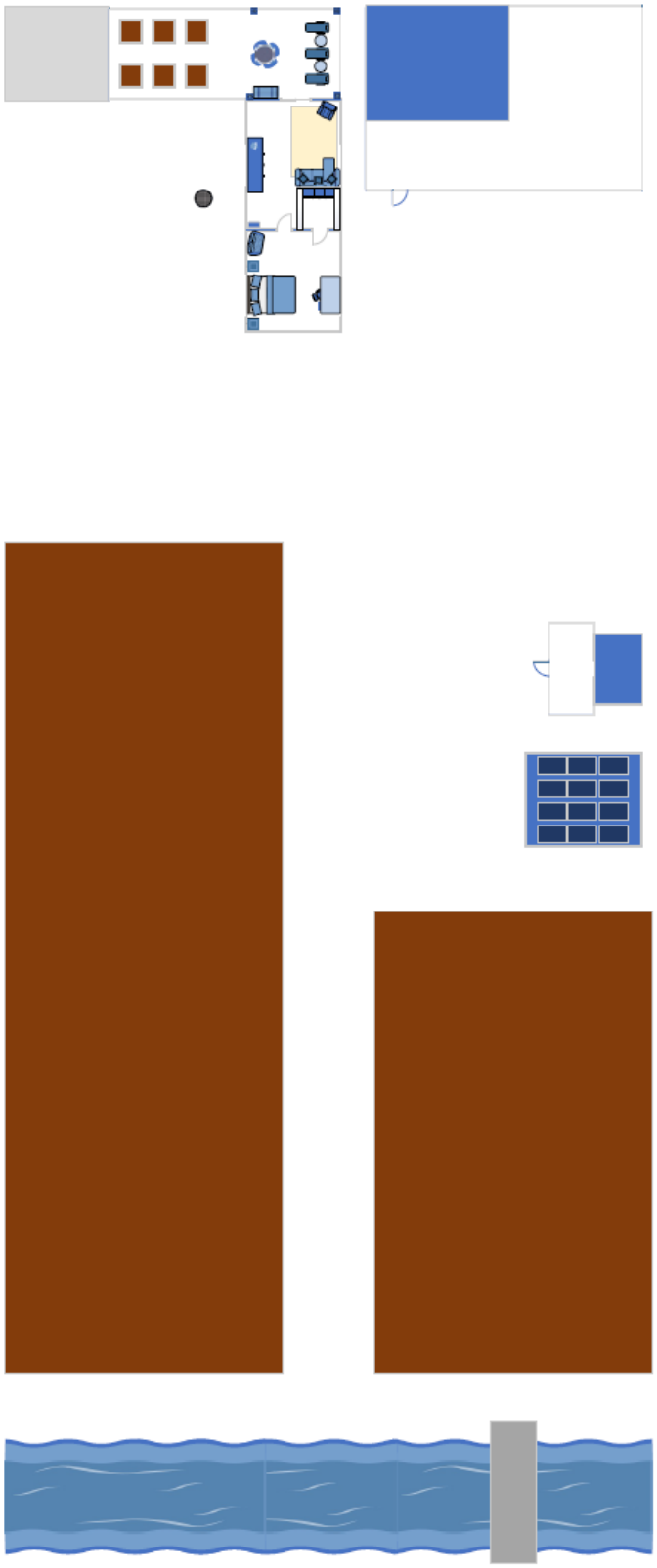


CAD drawings

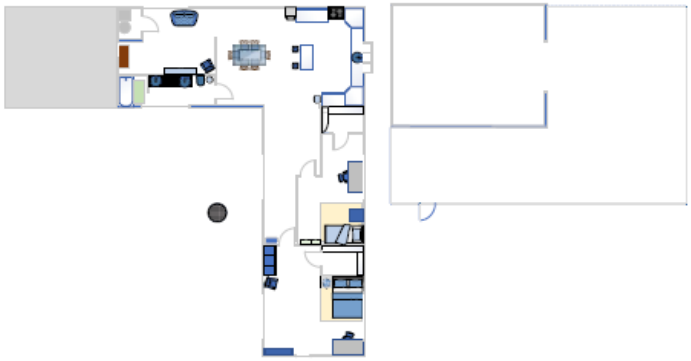


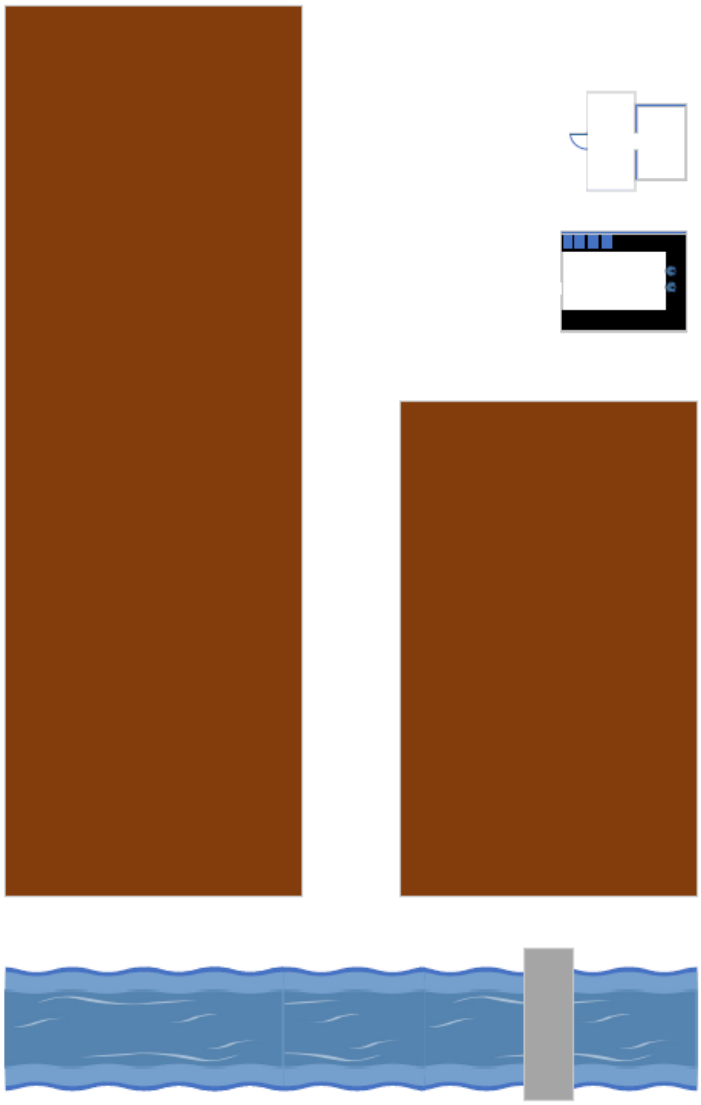


Skyview

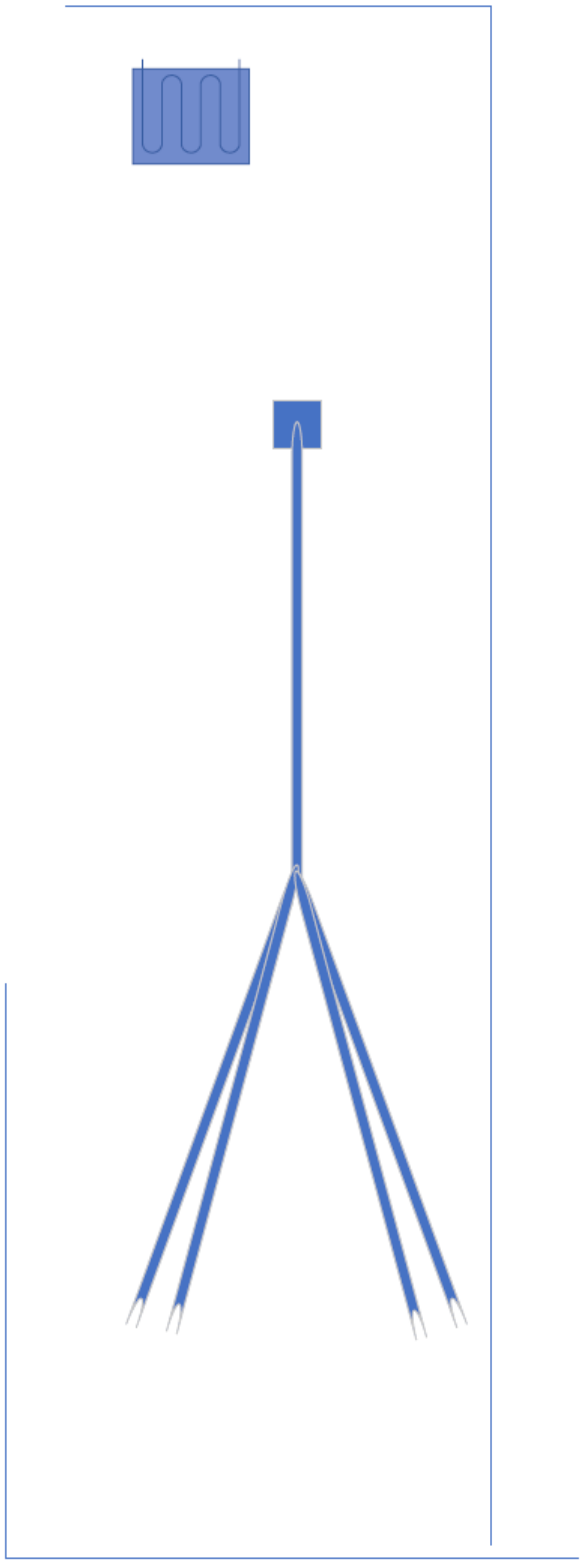


Second Floor





First floor



Below Ground

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Plan of Work Log

FORMS APPENDIX

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
25 Aug 2022 1.	Brainstorm	30 minutes	AS, JK, KJ, OL	Discussed possible countries
8 Sep 2022 2.	Brainstorm	1 hour	AS, JK, KJ, OL	Discussed possible locations in different countries
28 Sep 2022 3.	Brainstorm	30 minutes	AS, JK, KJ	Discussed ideas for home design
14 Oct 2022 4.	Divide Tasks	2 hours	AS, JK, KJ, OL	Divide research and plan other tasks
26 Oct 2022 5.	Research	30 minutes	AS, JK, KJ	Look up sourses and take notes
9 Nov 2022 6.	Research	30 minutes	AS, JK, KJ	Look up sourses and take notes

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature:

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
27 Nov 2022 1.	Home design	2 hours	AS, KJ	Work on CAD of home design
7 Dec 2022 2.	Research	30 minutes	AS, JK, KJ	Research and take notes
15 Dec 2022 3.	Research	1 hour	AS, KJ, JK, OL	Research and take notes
8 Jan 2023 4.	Write	2 hours	AS, KJ, JK, OL	Take research and begin writing portfolio
22 Jan 2023 5.	Brainstorm	30 minutes	AS, KJ, JK, OL	Think of materials to build model
29 Jan 2023 6.	Brainstorm	1 hour	AS, JK	Discuss plan for getting supplies and building model

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature: _____

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1 Feb 2023 1.	CAD	1 hour	KJ	Work on CAD of home and landscape
1 Feb 2023 2.	Research	30 minutes	OL	Continue assigned research
1 Feb 2023 3.	Model	1 hour	AS, JK	Start obtaining supplies for model
4 Feb 2023 4.	Model	2 hours	AS, JK	Discuss additional supplies needed and determine other remaining tasks
17 Feb 2023 5.	Portfolio	3 hours	AS, JK, KJ	Combine research into portfolio
21 Feb 2023 6.	Review	45 minutes	AS, JK, KJ, OL	Discuss remaining tasks

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature: _____

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
28 Feb 2023 1.	Review	1 hour	AS, JK, KJ, OL	Discuss needed additions to the portfolio
2 Mar 2023 2.	CAD/Model/ Portfolio	3 hours	AS, JK	Finalize CAD, obtain materials for model, proofread portfolio
4 Mar 2023 3.	Model	4 hours	AS	Build model
5 Mar 2023 4.	Model	2 hours	AS	Build model
6 Mar 2023 5.	Portfolio	2 hours	AS	Proofread and finalize portfolio
7 Mar 2023 6.	Portfolio	30 minutes	AS	Submit finished portfolio

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature: _____

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
1. 4 May 2023	Brainstorm	1.5 hours	AS, JK, KJ	Brainstormed ideas for refining the portfolio and model.
2. 13 May 2023	Portfolio	45 minutes	AS	Worked to update portfolio based on recommendations from state judges.
3. 16 May 2023	Model	2 hours	AS, JK, KJ, OL	Made plans for modifying model based on recommendations from state judges.
4. 8 Jun 2023	Model/Portfolio	1 hour	AS, JK, KJ	Made updates to portfolio and model
5. 12 Jun 2023	Portfolio	1 hour	AS, JK	Reviewed portfolio to identify additional information needed.
6. 15 Jun 2023	Model	3 hours	AS	Worked on designs to be 3D printed

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature: _____

TECHNOLOGY STUDENT ASSOCIATION PLAN OF WORK LOG

Date	Task	Time involved	Team member responsible (student initials)	Comments
19 Jun 2023 1.	Model	4 hours	AS	3D print items for model
21 Jun 2023 2.	CAD	4 hours	AS	Updated CAD
22 Jun 2023 3.	Model	1 hour	AS	Get final 3D prints for model
23 Jun 2023 4.	Model	3 hours	AS	Worked on model
24 Jun 2023 5.	Model/CAD	5 hours	AS	Finalized model and CAD
25 Jun 2023 6.	Portfolio	2 hours	AS	Proofread and made finial revisions to portfolio.

Advisor Name: [redacted] Student Initials: AS, JK, KJ, OL

Advisor Signature: _____